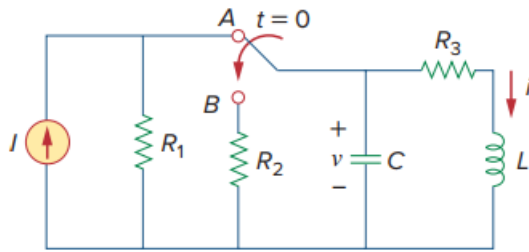


Problem

Using Fig. 8.100, design a problem to help other students better understand general second-order circuits.

Figure 8.100:



Step-by-step solution

Step 1 of 11

The given circuit is

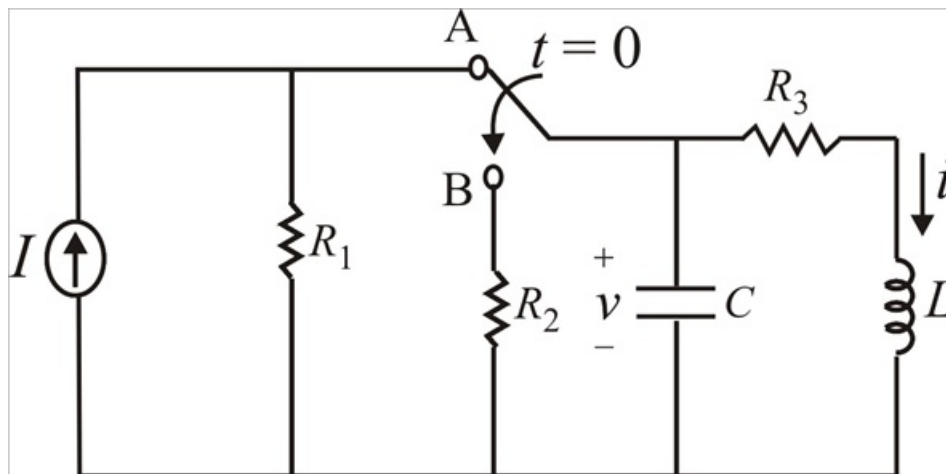


Fig-1

Step 2 of 11

For $t < 0$, the switch is at position A, and the circuit is in steady state, therefore the capacitor is replaced by an open circuit and the inductor by a short circuit,

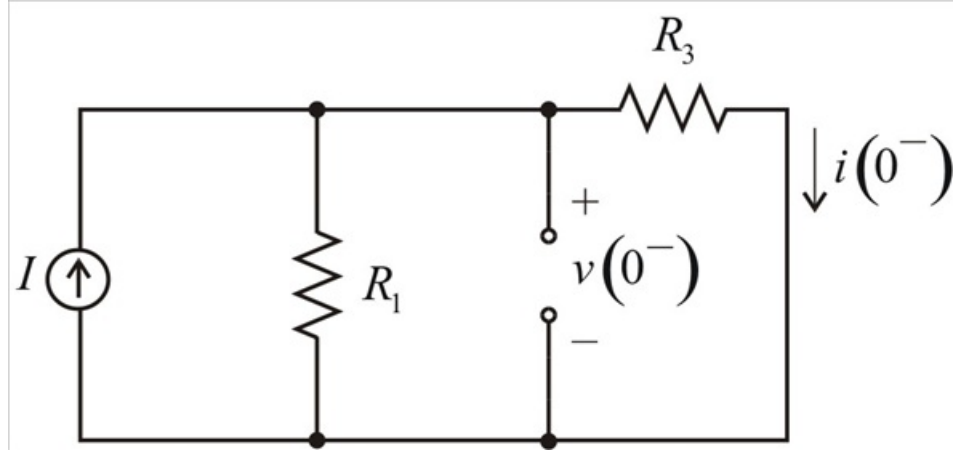


Fig-2

Step 3 of 11

The current through the inductor is given by current division rule,

$$i(0^-) = I \left[\frac{R_1}{R_1 + R_3} \right]$$

The voltage across the capacitor is,

$$v(0^-) = R_3 i(0^-)$$

$$v(0^-) = I \left[\frac{R_1 R_3}{R_1 + R_3} \right]$$

Step 4 of 11

Due to the continuity of the capacitor voltage and inductor current,

$$v(0^+) = v(0^-)$$

$$v(0^+) = I \left[\frac{R_1 R_3}{R_1 + R_3} \right]$$

$$i(0^+) = i(0^-)$$

$$i(0^+) = I \left[\frac{R_1}{R_1 + R_3} \right]$$

Step 5 of 11

For $t > 0$, the switch is moved to position B, therefore the equivalent circuit is,

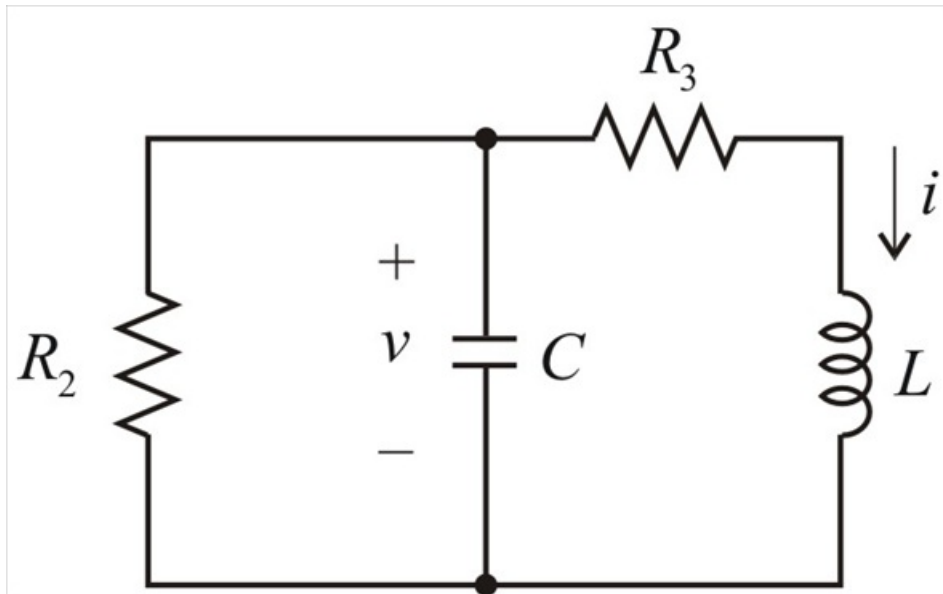


Fig-3

Step 6 of 11

Writing KCL at the top node, we get

$$\frac{v}{R_2} + C \frac{dv}{dt} + i = 0 \dots\dots\dots (1)$$

Applying KVL to the right side loop,

$$-v + R_3 i + L \frac{di}{dt} = 0$$

$$v = R_3 i + L \frac{di}{dt} \dots\dots\dots (2)$$

Step 7 of 11

Substituting v from equation (2) into (1), we get

$$\frac{R_3 i + L \frac{di}{dt}}{R_2} + C \frac{d}{dt} \left[R_3 i + L \frac{di}{dt} \right] + i = 0$$

$$\frac{R_3}{R_2} i + \frac{L}{R_2} \frac{di}{dt} + CR_3 \frac{di}{dt} + LC \frac{d^2 i}{dt^2} + i = 0$$

$$LC \frac{d^2 i}{dt^2} + \left(\frac{L}{R_2} + CR_3 \right) \frac{di}{dt} + \left(1 + \frac{R_3}{R_2} \right) i = 0$$

$$\frac{d^2 i}{dt^2} + \frac{1}{LC} \left(\frac{L}{R_2} + CR_3 \right) \frac{di}{dt} + \frac{1}{LC} \left(1 + \frac{R_3}{R_2} \right) i = 0$$

$$\frac{d^2 i}{dt^2} + \left(\frac{1}{R_2 C} + \frac{R_3}{L} \right) \frac{di}{dt} + \frac{1}{LC} \left(1 + \frac{R_3}{R_2} \right) i = 0$$

The characteristic equation is,

$$s^2 + \left(\frac{1}{R_2 C} + \frac{R_3}{L} \right) s + \frac{1}{LC} \left(1 + \frac{R_3}{R_2} \right) = 0$$

The roots are found by

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$$

There are three types of responses,

1. $\alpha > \omega_o$, over damped response (s_1, s_2 are real and different)

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

2. $\alpha = \omega_o$, critically damped response (s_1, s_2 are real and equal)

$$i(t) = (A_1 + A_2 t) e^{s t}$$

3. $\alpha < \omega_o$, under damped response (s_1 , and s_2 are complex)

$$i(t) = (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t}$$

Where $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$

Step 9 of 11

The constants A_1, A_2, B_1 and B_2 can be determined using the initial conditions.

The initial conditions are given by

$$i(0) = I \left[\frac{R_1}{R_1 + R_3} \right]$$

$$v(0) = I \left[\frac{R_1 R_3}{R_1 + R_3} \right]$$

Step 10 of 11

At $t = 0^+$,

From equation (2),

$$v(0^+) = R_3 i(0^+) + L \frac{di(0^+)}{dt}$$

$$\frac{di(0^+)}{dt} = \frac{v(0^+) - R_3 i(0^+)}{L}$$

Step 11 of 11

From equation (1),

$$\frac{v(0^+)}{R_2} + C \frac{dv(0^+)}{dt} + i(0^+) = 0$$

$$\frac{dv(0^+)}{dt} = \frac{\frac{-v(0^+)}{R_2} - i(0^+)}{C}$$

Substituting $i(t)$ in equation (2), we will get the voltage $v(t)$.